Chemistry: CYL1010

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References:
Organic Chemistry
Textbook by Jonathan Clayden, Nick Greeves, and Stuart Warren

Fundamentals of Organic Chemistry
Book by T. W. Graham Solomons

Transition Metal Chemistry [5 Lectures]: Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry

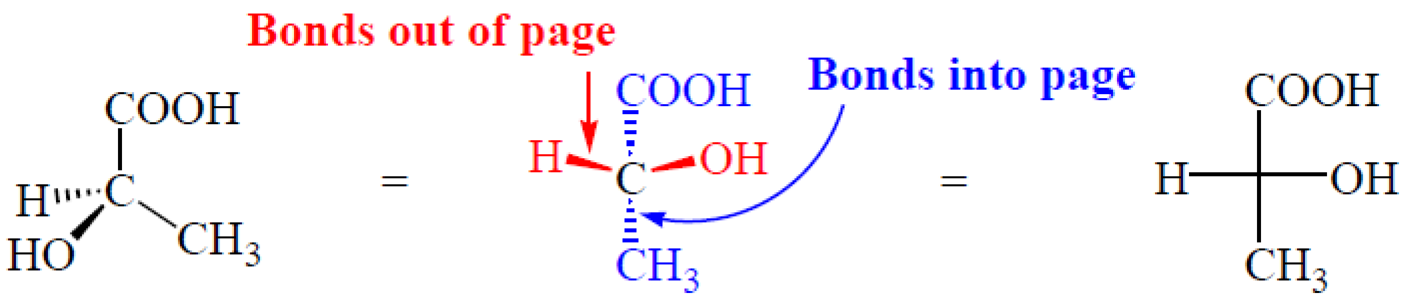
Organic Chemistry [8 Lectures]: Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies

Fischer Projection

1. The carbon chain is drawn along the vertical line of the Fischer projection, usually with the most highly oxidized end carbon atom at the top.

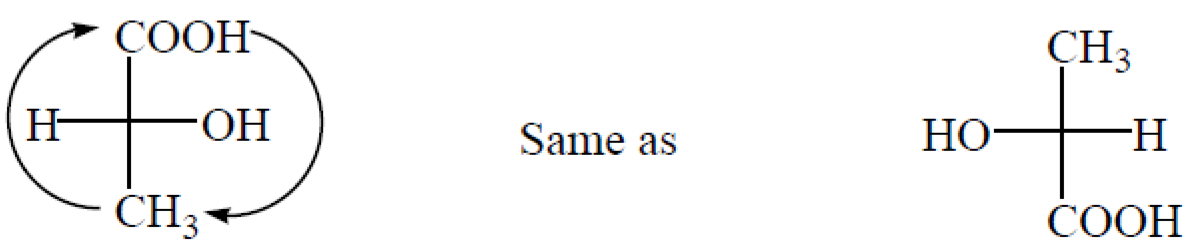
Vertical lines: bonds going into the page.

Horizontal lines: bonds coming out of the page

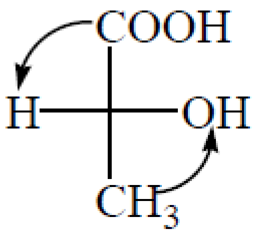
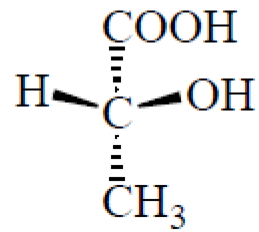


(*R*)-Lactic acid **Fischer projection**

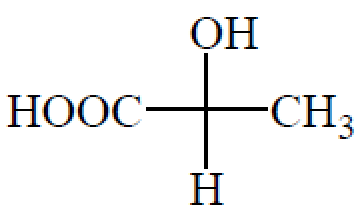
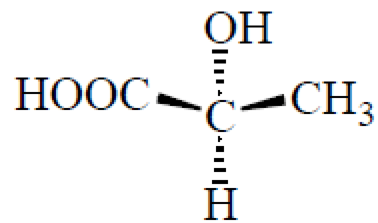
180° rotation (not 90° or 270°):



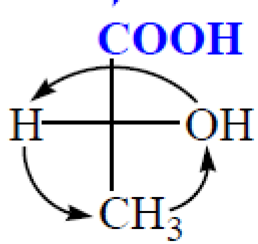
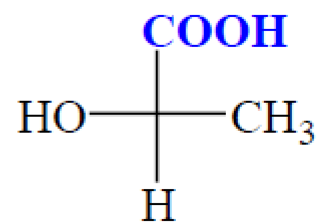
–COOH and –CH_3 go into plane of paper in both projections;
–H and –OH come out of plane of paper in both projections.

 Same as  (***R***)-**Lactic acid**

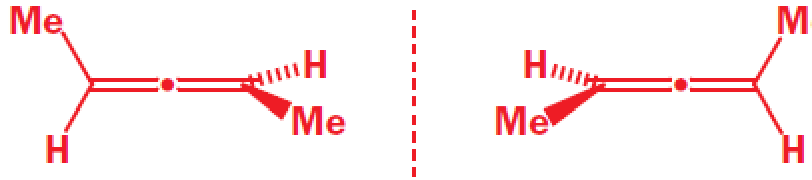
$\downarrow$ 90° rotation

 Same as  (***S***)-**Lactic acid**

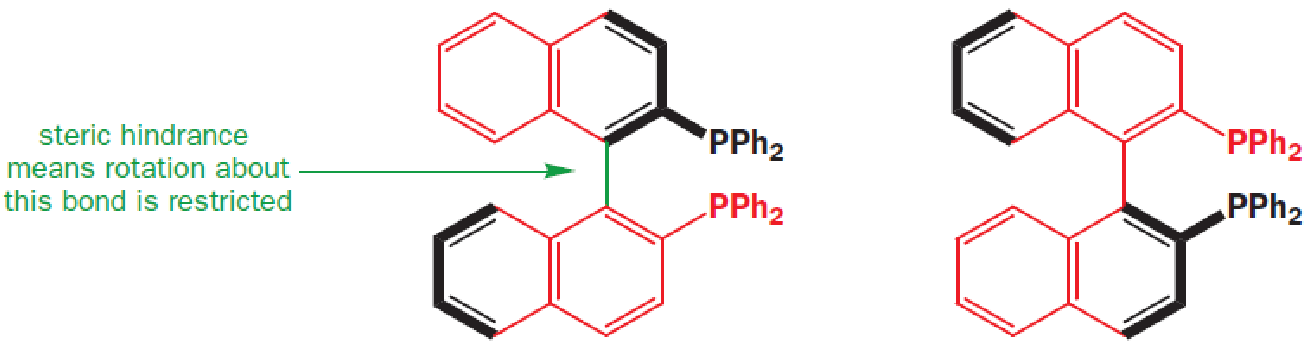
Hold steady

 Same as 

Chiral compounds with no stereogenic centres

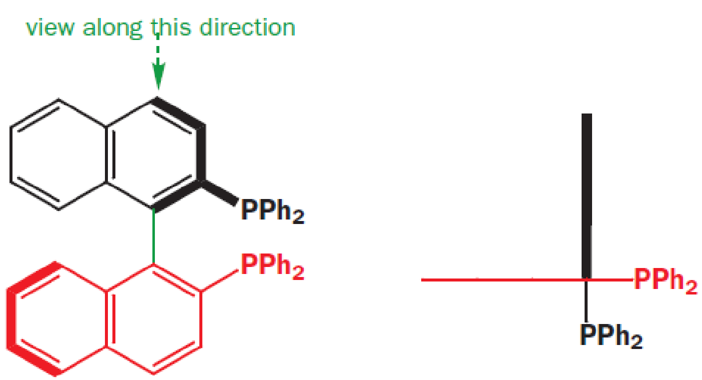


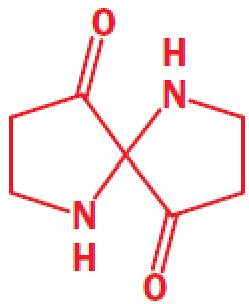
a chiral allene

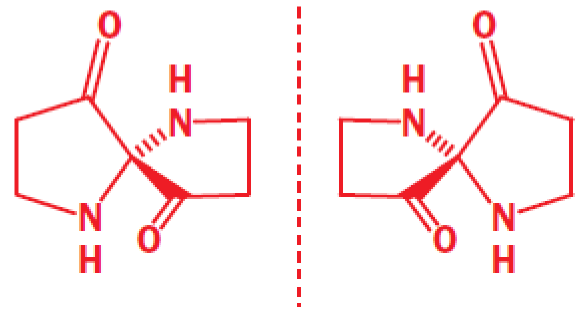


(*R*)-BINAP

(*S*)-BINAP







nonsuperimposable enantiomers

Resolution

Separation of Enantiomers

1. Resolution *via* Diastereomer Formation:

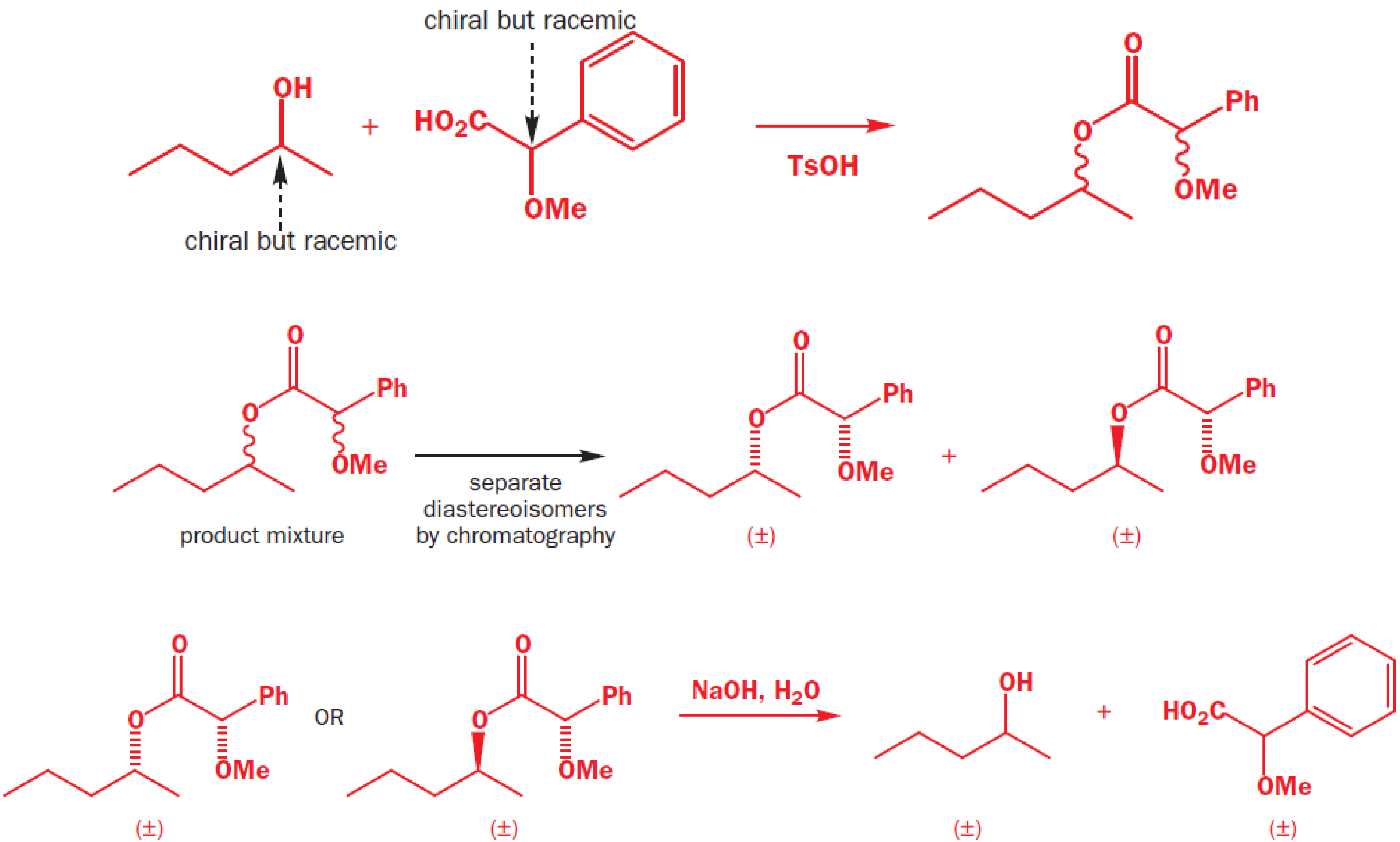
- Diastereomers, because they have different melting points, different boiling points, and different solubilities, can be separated by conventional methods.

2. Resolution *via* Molecular Complexes, Metal Complexes, and Inclusion Compounds.

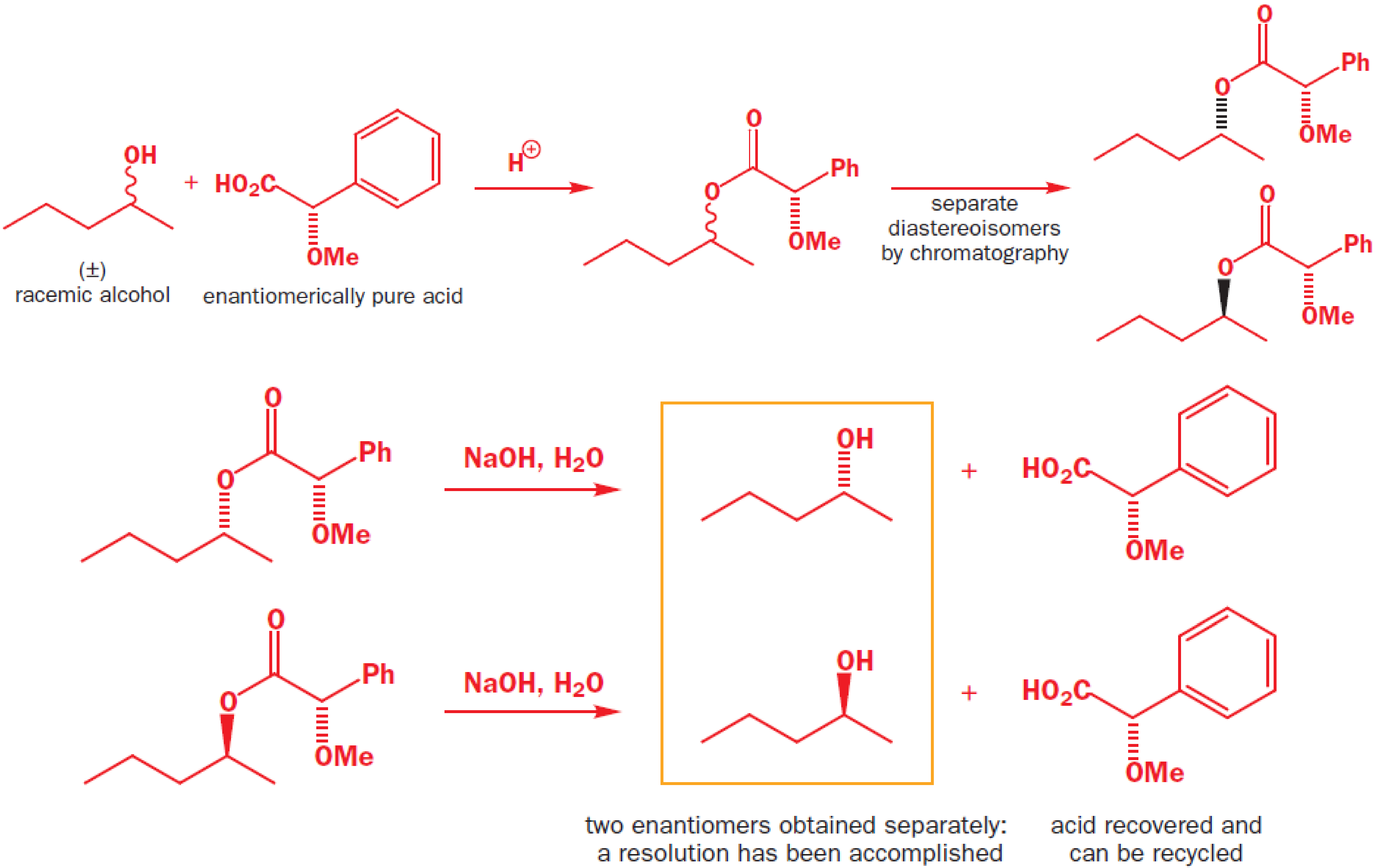
3. Chromatographic Resolution

4. Kinetic Resolution

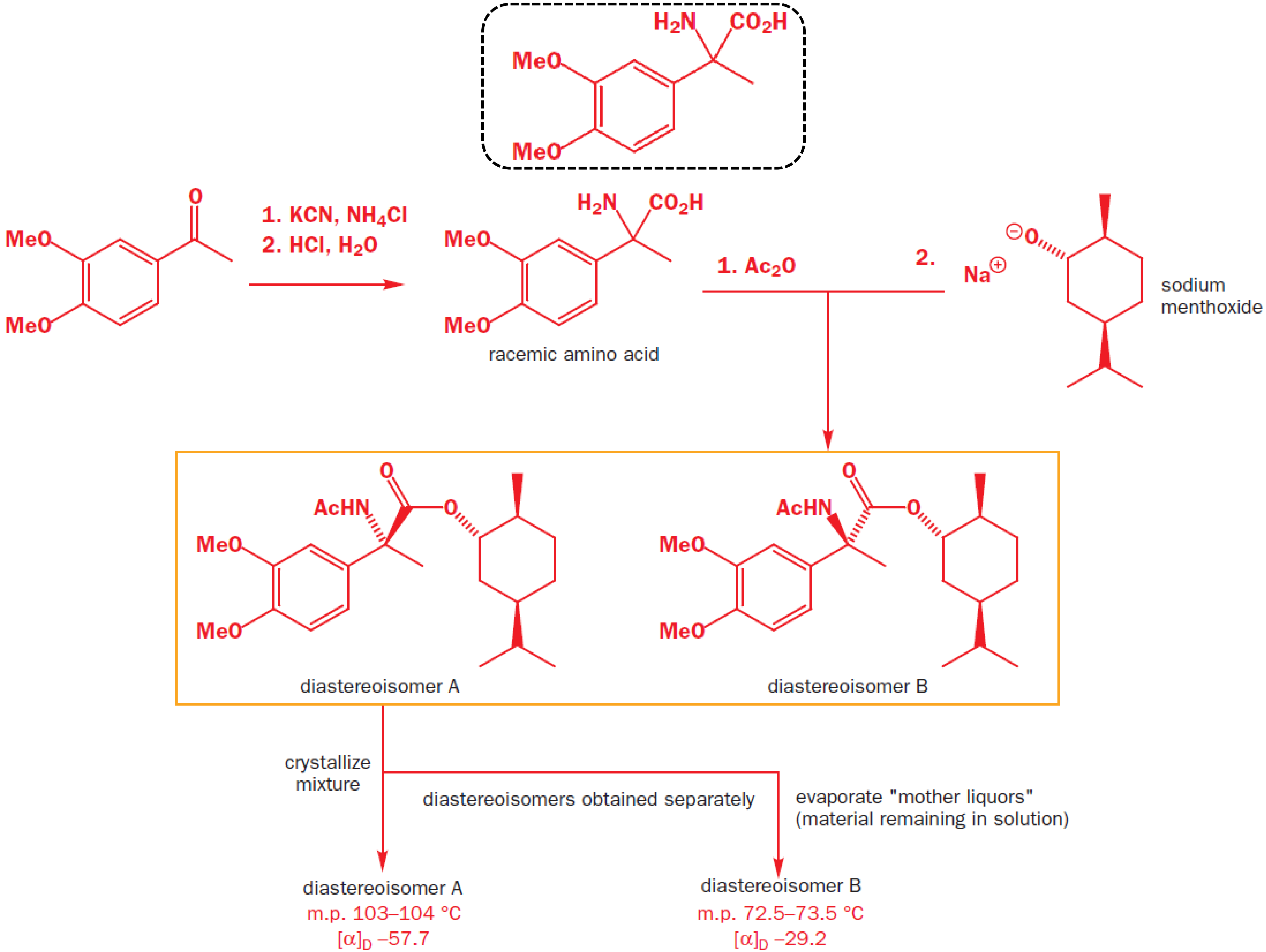
Resolution

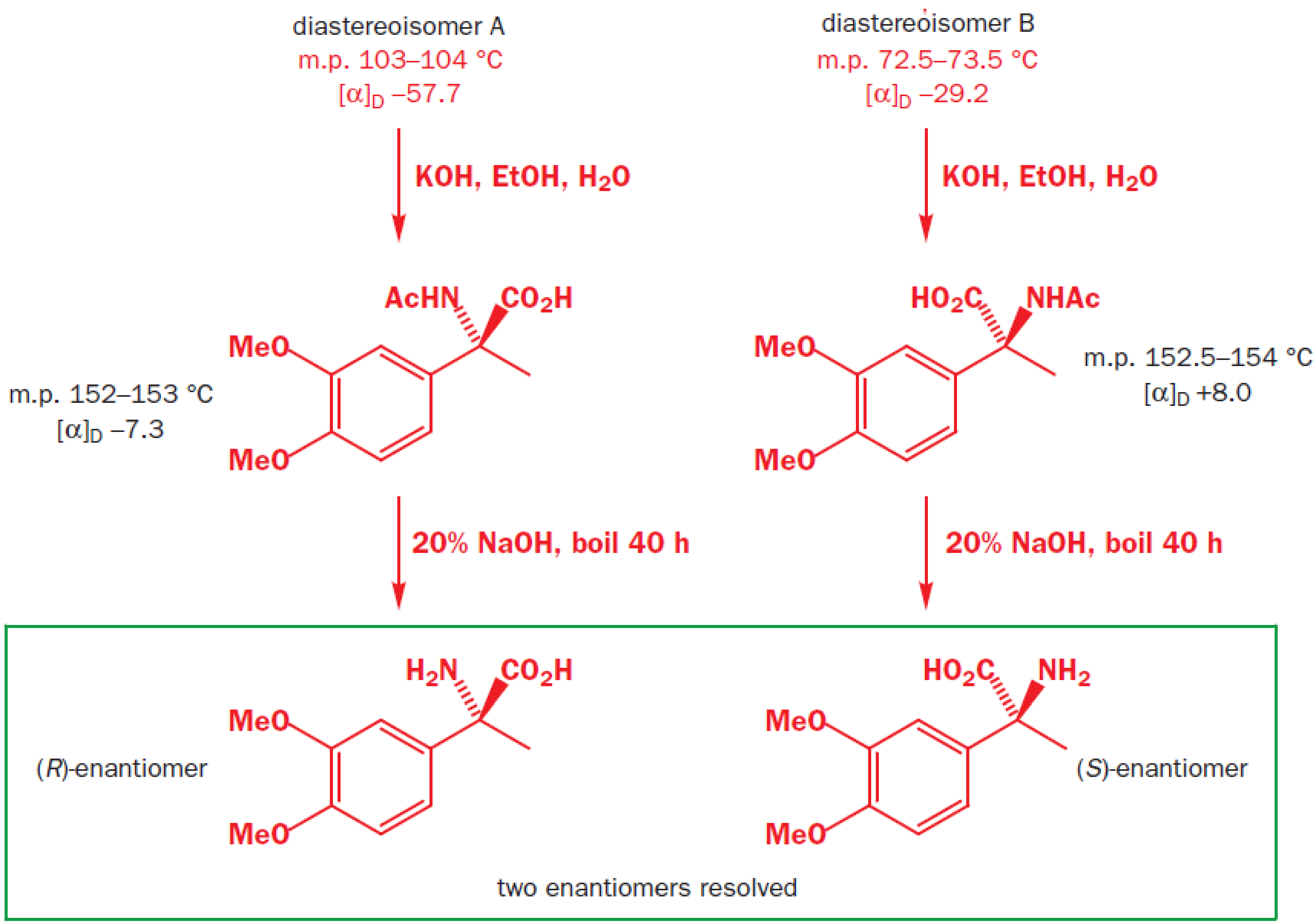


chiral but racemic
OH
+
HO2C
OMe
chiral but racemic
TsOH
O
Ph
OMe
product mixture
separate
diastereoisomers
by chromatography
(±)
+
(±)
OR
NaOH, H2O
(±)

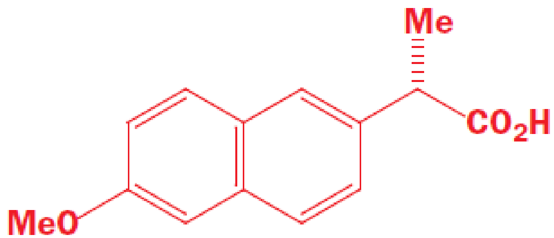


A separation of two enantiomers is called a **resolution**. Resolutions can be carried out only if we make use of a component that is already enantiomerically pure.

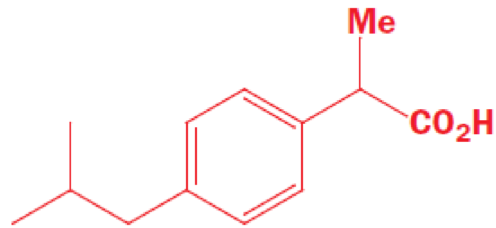

H2N
CO2H
MeO
MeO
O
1. KCN, NH4Cl
2. HCl, H2O
racemic amino acid
1. Ac2O
2.
Na⊕
⊖O
sodium menthoxide
AcHN
diastereoisomer A
diastereoisomer B
crystallize mixture
diastereoisomers obtained separately
evaporate "mother liquors" (material remaining in solution)
diastereoisomer A
m.p. 103–104 °C
[α]D –57.7
diastereoisomer B
m.p. 72.5–73.5 °C
[α]D –29.2


diastereoisomer A
m.p. 103–104 °C
[α]D –57.7
KOH, EtOH, H2O
AcHN
CO2H
MeO
MeO
m.p. 152–153 °C
[α]D –7.3
20% NaOH, boil 40 h
diastereoisomer B
m.p. 72.5–73.5 °C
[α]D –29.2
KOH, EtOH, H2O
HO2C
NHAc
MeO
MeO
m.p. 152.5–154 °C
[α]D +8.0
20% NaOH, boil 40 h
H2N
CO2H
MeO
MeO
(R)-enantiomer
HO2C
NH2
MeO
MeO
(S)-enantiomer
two enantiomers resolved

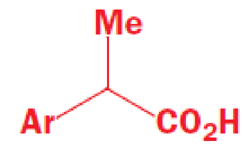
Resolutions using diastereoisomeric salts



(*S*)-naproxen

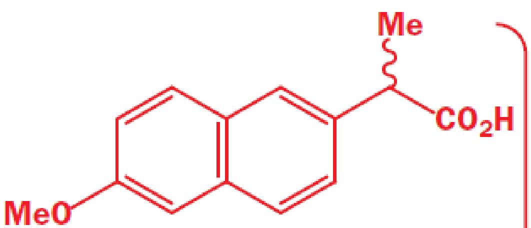


ibuprofen

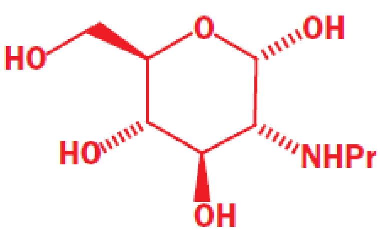


2-arylpropionic acids

resolution of naproxen via an amine salt

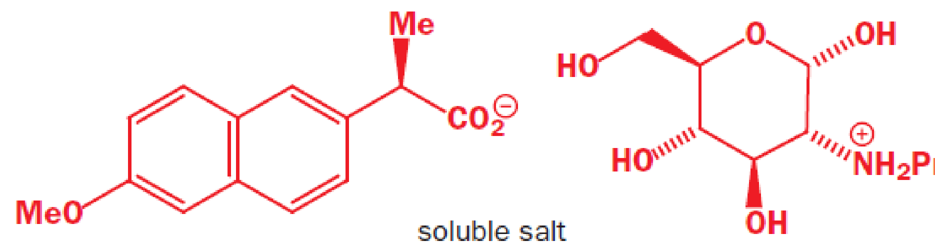


(±)-naproxen

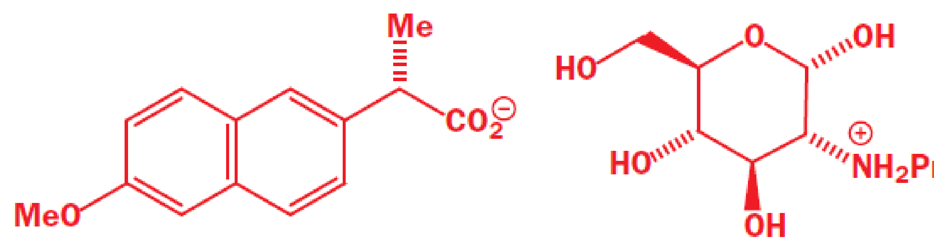


N-propylglucosamine

mixed to give

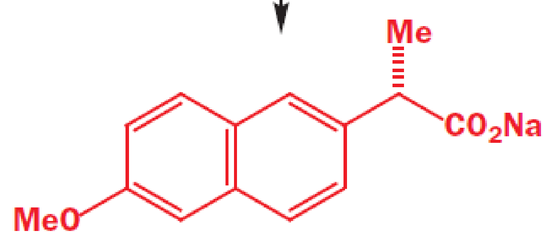


soluble salt



crystalline salt

filtered off and
basified with NaOH

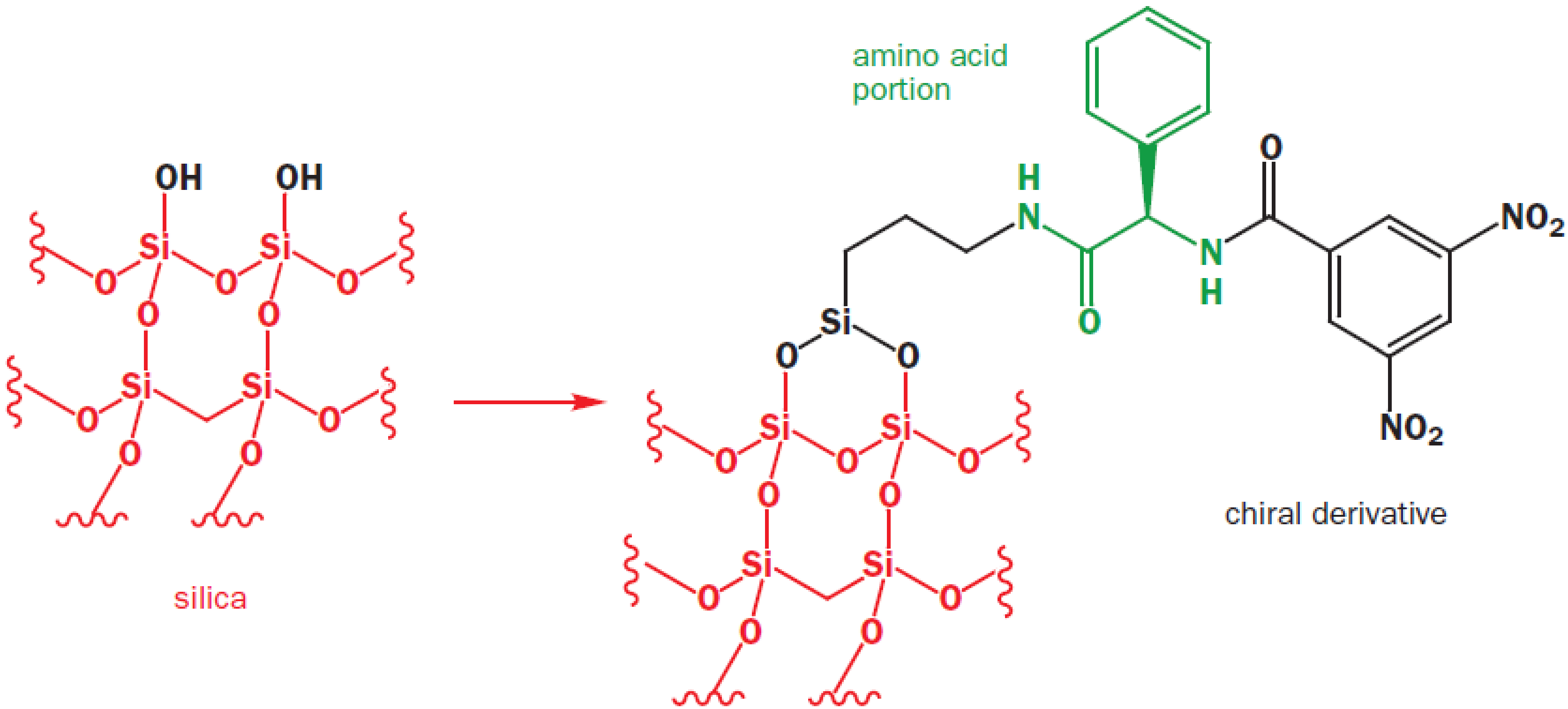


sodium salt of (*S*)-naproxen

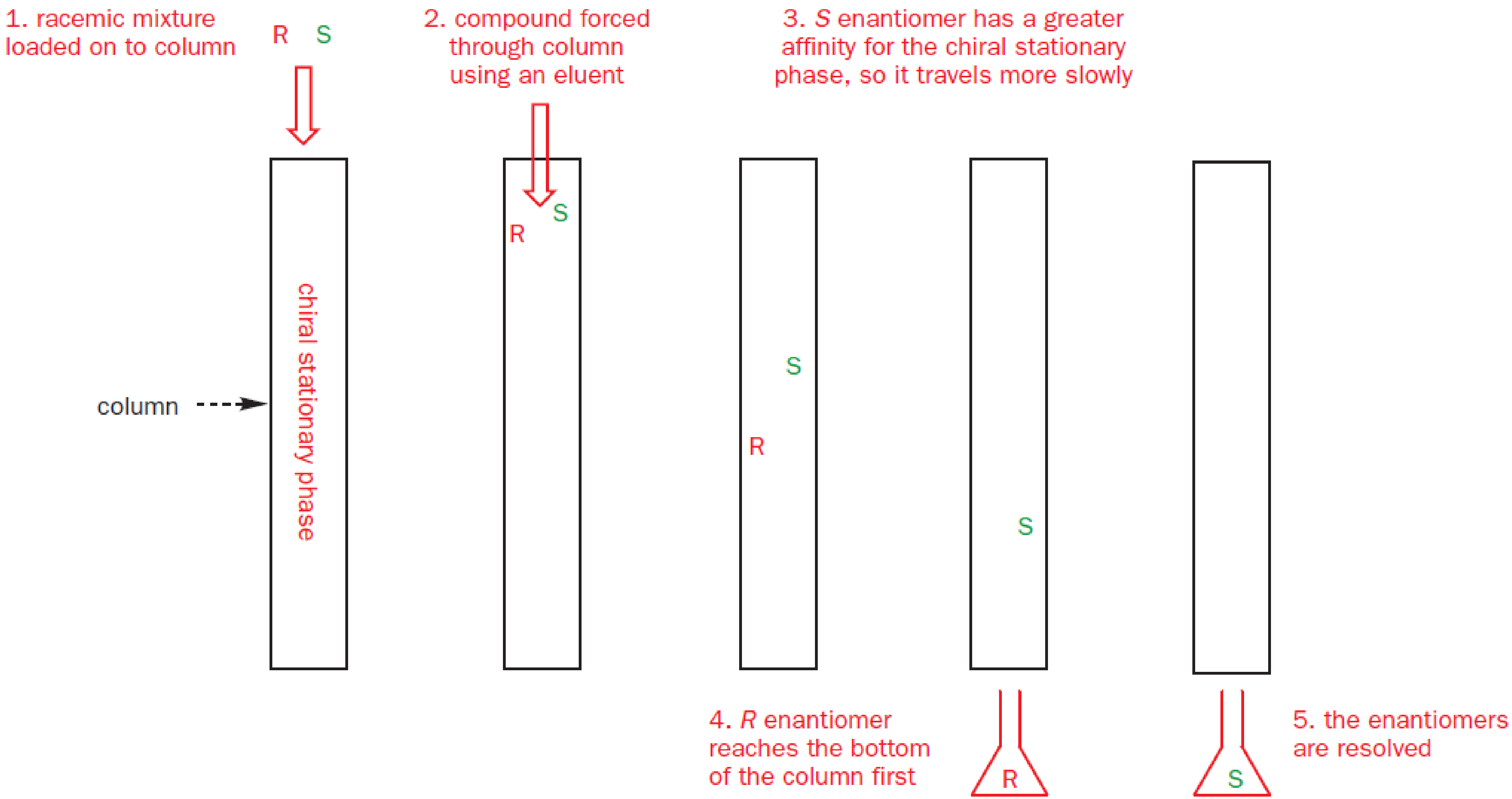
reusable

N-propylglucosamine plus

Resolution by Column Chromatography



Chromatography on a chiral stationary phase is especially important when the compounds being resolved have no functional groups suitable for making the derivatives (usually esters or salts) needed for the more classical resolutions.



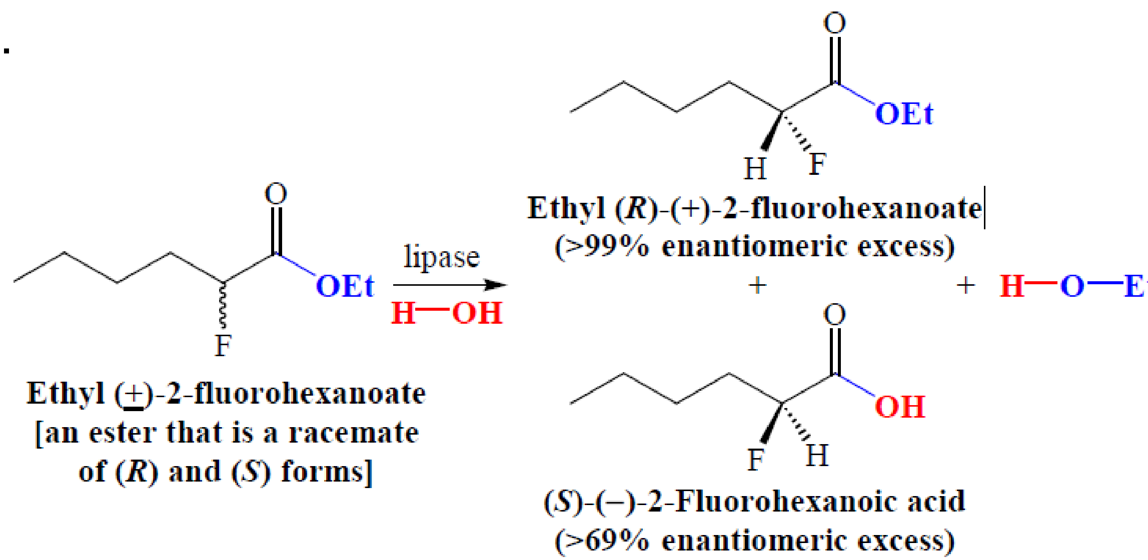
1. racemic mixture loaded on to column
R S
2. compound forced through column using an eluent
S
R
3. *S* enantiomer has a greater affinity for the chiral stationary phase, so it travels more slowly
S
R
S
column
chiral stationary phase
4. *R* enantiomer reaches the bottom of the column first
R
5. the enantiomers are resolved
S

Enantioselective Synthesis

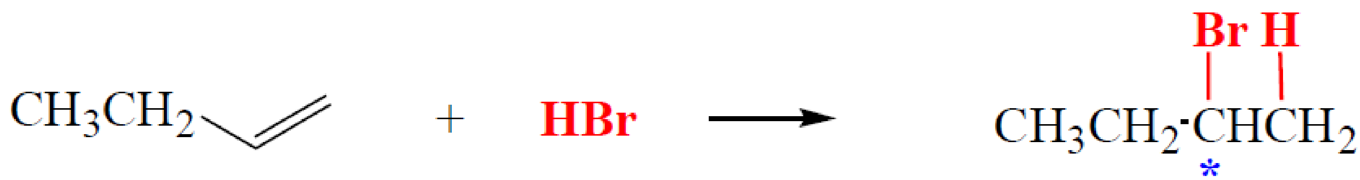
1) In an enantioselective reaction, one enantiomer is produced predominantly over its mirror image.

2) In an enantioselective reaction, a chiral reagent, catalyst, or solvent must assert an influence on the course of the reaction.

For example, in nature, where most reactions are enantioselective, the chiral influences come from protein molecules called enzymes. Enzymes possess an *active site* where only one enantiomer of a chiral reactant fits it properly and is able to undergo reaction.

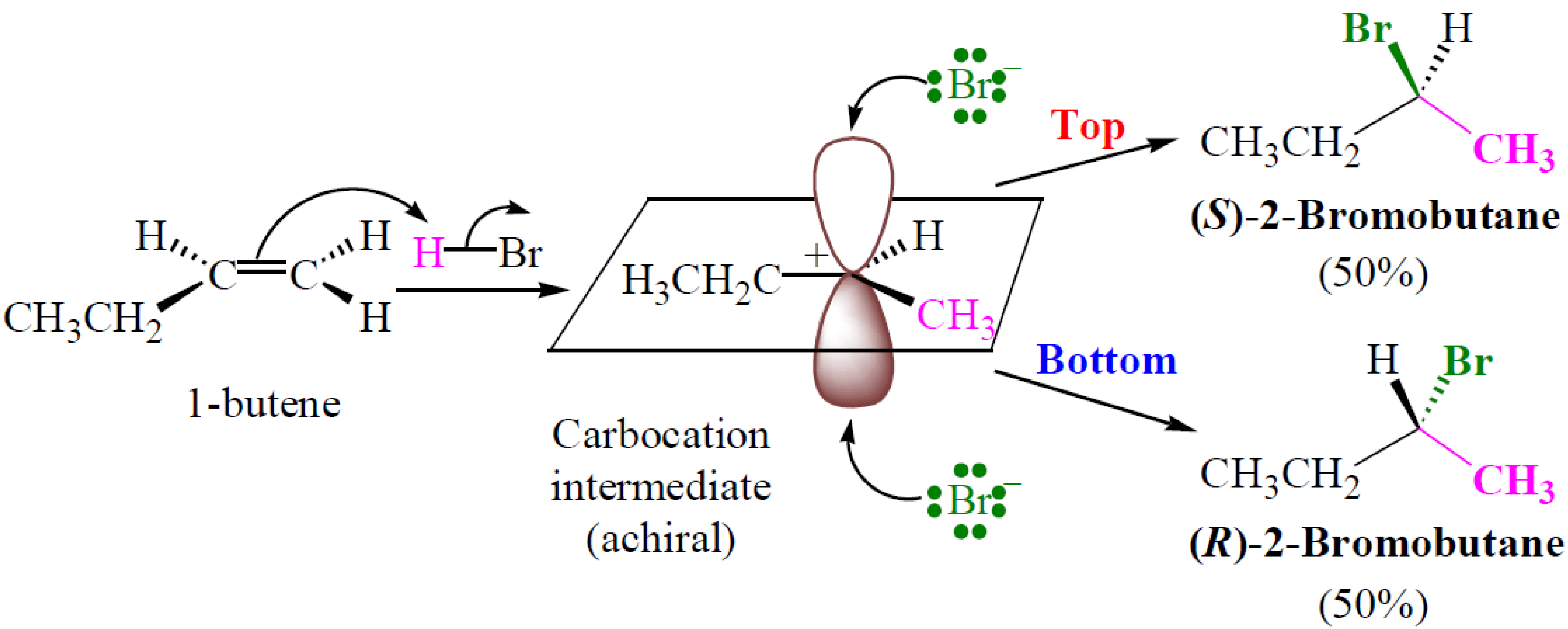


Racemic Synthesis



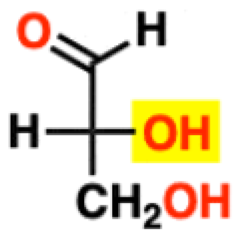
1-Butene
(chiral)

(±)-2-Bromobutane
(achiral)

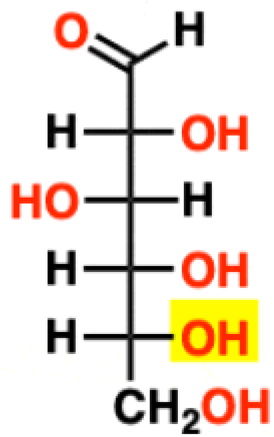


For a sugar drawn in the Fischer projection with the most oxidized carbon at the top:

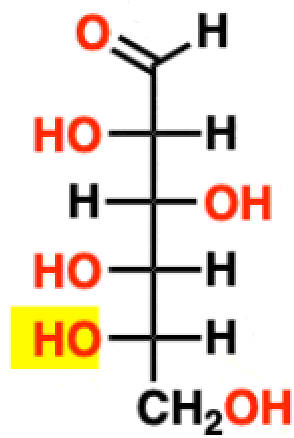
- If the OH on the bottom chiral center points to the **right**, the sugar is **D**
- If the OH on the bottom chiral center points to the **left**, the sugar is **L**



D-glyceraldehyde

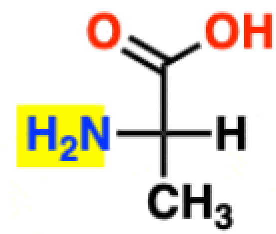


D-Glucose

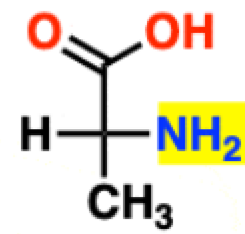


L-Glucose

The D- L- system can also be applied to other chiral molecules, e.g. amino acids:



L-alanine



D-alanine